BM25 on FiQA: parameter tuning and comparison

Yana Malashkina · 9 November 2026 · 02456 retrieval project

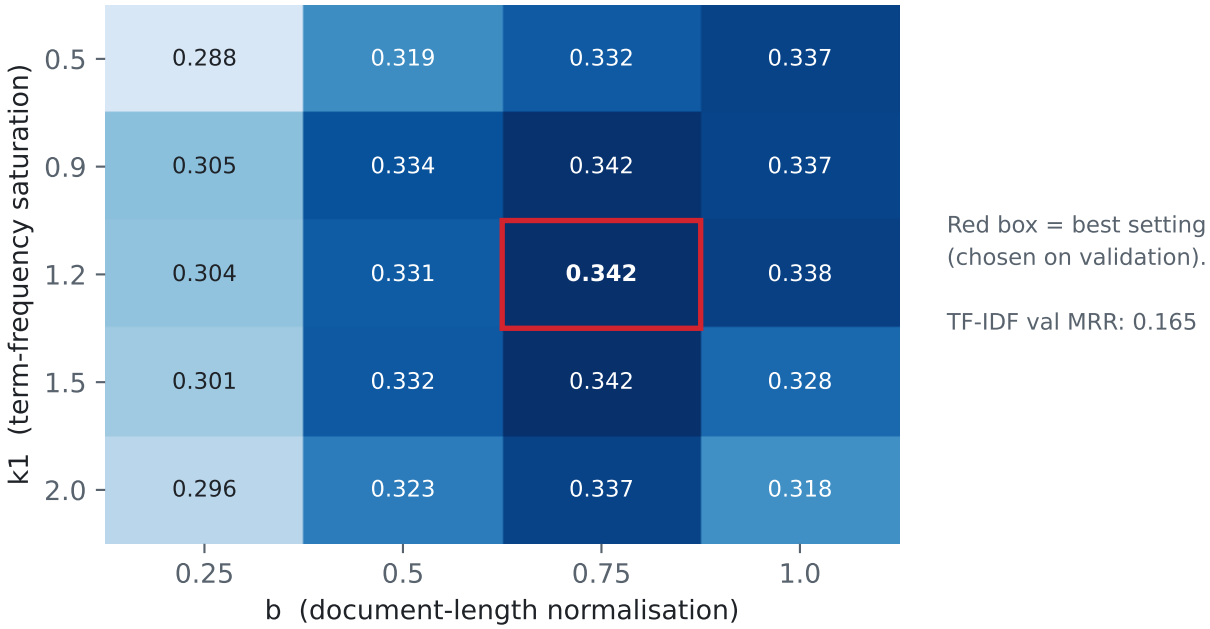
Setup

Corpus: 17,071 FiQA documents. Queries: 500 validation, 648 test. Metrics at top-k = 10.
Grid search over k1 ∈ {0.5, 0.9, 1.2, 1.5, 2.0} and b ∈ {0.25, 0.5, 0.75, 1.0} (20 settings), selected by validation MRR@10. Test set used only for the final comparison.
Uses the project TF-IDF / BM25 implementations unchanged.

Results

Method	Val recall	Val MRR	Test recall	Test precision	Test MRR
TF-IDF	0.174	0.165	0.164	0.042	0.159
BM25 default (k1=1.5, b=0.75)	0.332	0.342	0.344	0.078	0.348
BM25 tuned (k1=1.2, b=0.75)	0.336	0.342	0.346	0.078	0.346

Validation MRR@10 for each (k1, b)



Key findings

- BM25 is more than twice as good as TF-IDF on test: MRR 0.348 vs 0.159 and recall@10 0.344 vs 0.164.
- b (length normalisation) matters most: b = 0.25 is clearly worst (mean val MRR 0.299), b = 0.75 is best (0.339).
- k1 matters much less: at b = 0.75, all k1 values from 0.5 to 2.0 give val MRR between 0.332 and 0.342.
- The default k1 = 1.5, b = 0.75 is already near-optimal. Tuning picked k1 = 1.2 (val MRR 0.3421 vs 0.3420), but on test it is equal, not better (recall 0.346 vs 0.344, MRR 0.346 vs 0.348).
- Precision@10 is low (~0.08) because most queries have only a few relevant documents, so 10 results cannot all be relevant.

Reproduce: grid search with bm25_experiments.py; a single setting with
python main.py retriever=bm25 retriever.k1=<k1> retriever.b=<b>